School Choice, Deferred Acceptance, and Student Outcomes: Evidence from a Centralized Admissions System

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Abstract

We study the welfare effects of switching from the Boston Immediate Acceptance mechanism to the Deferred Acceptance (DA) algorithm for school assignment in a large urban district. Using administrative data on 155,000 applicants across fourteen cohorts, we document that DA assigns 66 percent of students to their first-choice school – 5 percentage points below the Boston mechanism under truthful play, but 7 percentage points above the Boston mechanism in Nash equilibrium with strategic behavior. Using regression discontinuity designs at school admission cutoffs, we estimate that attending an oversubscribed school improves 8th grade math scores by 0.30 standard deviations, reading by 0.20 standard deviations, high school graduation by 6.9 percentage points, and four-year college enrollment by 5.1 percentage points. Effects are largest for low-income students and English language learners. Counterfactual welfare analysis shows that DA achieves Pareto improvements over Boston for 12 percent of students, and that Top Trading Cycles achieves higher average welfare but at the cost of stability and justified envy.

Contents

1. Introduction

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on market design by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in market design is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and

falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

Our paper relates to several strands of the economics literature. First, we contribute to the extensive body of work on the determinants of the outcome variable, which has been a central topic in market design since the seminal contributions of the 1990s and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third, our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the

outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

Before proceeding, we note several contributions relative to the closest existing work. While prior studies have documented a correlation between the treatment variable and the outcome variable in various contexts, our paper is among the first to provide credibly causal estimates using an instrumental variables strategy. Moreover, our analysis of mechanisms goes beyond what has been possible in earlier studies, shedding light on the channels through which the treatment variable affects the outcome variable.

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2..3 Data Sources and Sample Construction

Our primary data source is administrative records, which covers 50,000 individuals over the period 1990–2010. This dataset provides detailed information on the outcome variable, the treatment variable, and a rich set of control variables that we use to address potential confounders. The data are available at the individual level and include both repeated cross-sections and a longitudinal component that allows us to track individuals over time. We discuss the advantages and limitations of this dataset relative to alternatives in the literature.

Table **??** presents summary statistics for the key variables used in our analysis. The mean of the outcome variable is consistent with values reported in prior studies, providing confidence that our sample is representative of the population of interest. The standard deviation suggests substantial variation across individuals, which is essential for our identification strategy. We also report the 10th and 90th percentiles to characterize the full distribution and to assess the extent to which our results might be driven by outliers in the tails.

We construct our main sample by merging several administrative and survey datasets. After

applying standard sample restrictions and dropping observations with missing values on key variables, our final estimation sample contains 50,000 observations covering 1990–2010. The merge rate across datasets exceeds 95 percent, alleviating concerns about selective attrition at the matching stage. We verify that the merged sample is representative by comparing moments of the key variables to those in the underlying source datasets.

The key independent variable, the treatment variable, exhibits substantial cross-sectional variation. The interquartile range spans a factor of three, providing sufficient variation to identify the effect of interest. We verify that this variation is not driven by outliers or measurement error by conducting a series of diagnostic tests. In particular, we show that the results are robust to winsorizing the treatment variable at the 1st and 99th percentiles, to using alternative measures from different data sources, and to correcting for measurement error using an errors-in-variables estimator.

We supplement the main dataset with auxiliary sources to construct additional control variables and instruments. Geographic controls come from standard sources in the literature, including distance to the coast, latitude, rainfall, temperature, and terrain ruggedness. Demographic controls are measured at baseline to avoid bad control problems. Institutional controls are drawn from widely used cross-country datasets. We lag all time-varying controls by one period to mitigate simultaneity concerns.

The measurement of the outcome variable deserves discussion. Following the standard approach in the literature, we use the logarithm of the outcome variable as our primary outcome variable. This transformation has three advantages: it reduces the influence of outliers, it allows us to interpret coefficients as approximate percentage changes, and it is consistent with the multiplicative structure of the theoretical model. We verify that our results are qualitatively robust to using the outcome variable in levels or in ranks.

Panel A of Table **??** reports summary statistics for the full sample, while Panels B and C restrict attention to subsamples that will be used in the heterogeneity analysis. The key variables are well-balanced across subsamples, with the exception of the treatment variable, which exhibits the expected variation driven by the quasi-experimental design. We formally test for balance across subsamples using a multivariate F-test and fail to reject the null of joint equality at conventional significance levels.

Table 1: Student Summary Statistics

	All students	DA assigned	Not DA assigned	Difference
Female (%)	49.8	50.2	49.4	0.8
Free/reduced lunch (%)	62.4	64.1	60.8	3.3
Special education (%)	14.2	13.8	14.6	-0.8
English language learner (%)	18.7	19.4	18.1	1.3
4th grade math score (std. dev.)	0.00	0.04	-0.04	0.08
4th grade reading score (std. dev.)	0.00	0.02	-0.02	0.04
Age at application (months)	131.4	131.6	131.2	0.4

Notes: Sample: all applicants to the centralized school choice system, 2005-2018 cohorts. DA assigned = student was assigned to their first-choice school via DA. Differences are raw means differences; none are statistically significant at the 5% level after Bonferroni correction.

Table 2: School Characteristics

	All schools	Oversubscribed	Undersubscribed	Difference
Total enrollment	482	641	384	257***
Free/reduced lunch (%)	59.4	52.1	63.8	-11.7***
4th-8th grade test score (std. dev.)	0.00	0.41	-0.24	0.65***
Pupil-teacher ratio	16.4	15.8	16.8	-1.0**
Share of teachers with > 5 years experience (%)	61.4	68.4	57.1	11.3***
Distance from median applicant (km)	2.84	3.12	2.68	0.44

Notes: *** p<0.01, ** p<0.05, * p<0.1. Oversubscribed = more applicants than seats in at least one year. School-level mean weighted by enrollment.

Data quality is a potential concern in studies of this type. We address this concern in several ways. First, we cross-validate our primary measure of the outcome variable against alternative sources and find high concordance rates. Second, we examine the time series of key variables for breaks or anomalies that might indicate measurement problems. Third, we re-estimate our main specifications on a subsample that excludes observations flagged as potentially problematic by the data providers. All results are robust to these checks.

3. Matching Theory

Consider an economy populated by a continuum of agents indexed by $i \in [0, 1]$. Each agent chooses the outcome variable to maximize lifetime utility subject to a budget constraint. The key source of heterogeneity across agents is their exposure to the treatment variable. We solve for the equilibrium of this economy and derive comparative statics with respect to the key parameters. The model generates a clear testable prediction: an exogenous increase in the treatment variable should lead to a positive and significant change in the outcome variable, with the magnitude depending on structural parameters that we estimate in the empirical analysis.

Let Y_i denote the outcome of interest for unit i. We model Y_i as a function of treatment status D_i, observed covariates X_i, and unobserved heterogeneity ε_i. Under standard regularity conditions, the conditional expectation function takes the form presented in Equation **??**. The parameter of interest is the coefficient on treatment status, which captures the causal effect of the treatment variable on the outcome variable holding constant observed and unobserved confounders. Identification of this parameter requires an exclusion restriction, which we discuss in detail below.

The theoretical framework builds on the canonical model of field. Agents face a discrete choice between J alternatives, each yielding utility $U_{ij} = V_{ij} + \epsilon_{ij}$, where V_{ij} is the deterministic component and ϵ_{ij} captures idiosyncratic preferences. Under standard distributional assumptions on the error term, the choice probabilities take a tractable closed form. The model can be estimated using maximum likelihood or method of moments, and the resulting structural parameters can be used to simulate counterfactual policy scenarios.

We derive the key prediction of the model through a simple comparative statics exercise. An exogenous increase in the treatment variable shifts the equilibrium allocation of the outcome

variable through two channels: a direct effect operating through prices, and an indirect effect operating through the reallocation of factors across sectors. The total effect is ambiguous in general, but under the empirically relevant parameter values, we show that the direct effect dominates. Proposition 1 in Appendix A formalizes this result and establishes conditions under which the sign of the total effect is unambiguous.

Under the assumption that the proposed instrument satisfies the exclusion restriction, the model implies that the reduced-form relationship between the proposed instrument and the outcome variable is proportional to the structural parameter of interest. This motivates our instrumental variables strategy described in Section **??**. A key advantage of the IV approach is that it does not require functional form assumptions on the relationship between the treatment variable and the unobserved confounders. The cost is that we identify a local average treatment effect rather than the population average treatment effect.

$$\text{Step } t: \text{ each unmatched student } i \text{ proposes to best unrejected school } s; \text{ school } s \text{ tentatively holds top-}q_s \text{ and reje} \tag{1}$$

$$(i, s) \text{ is a blocking pair if } s \succ_i \mu(i) \text{ and } i \succ_s j \text{ for some } j \in \mu(s), \quad |\mathcal{S}| \leq \binom{n}{2} \cdot \lfloor K/2 \rfloor \tag{2}$$

$$\mu_D(\succ_i, \succ_{-i}) \succsim_i \mu_D(\succ_i', \succ_{-i}) \quad \forall \succ_i', \succ_{-i}, i \tag{3}$$

The model delivers additional testable implications beyond the main prediction. First, the treatment effect should be larger for individuals with lower baseline levels of the outcome variable, reflecting diminishing marginal returns. Second, the effect should be persistent over time rather than transitory, because the mechanism operates through permanent changes in the level of the treatment variable. Third, the effect should be concentrated in sectors with greater exposure to the underlying channel. We test each of these auxiliary predictions in Section 6.

We make several simplifying assumptions to maintain tractability. First, we assume that agents have rational expectations and perfect information about the distribution of shocks. Second, we abstract from general equilibrium effects by treating prices as exogenous to individual agents.

Third, we assume that the production technology exhibits constant returns to scale. We discuss the robustness of our theoretical predictions to relaxing each of these assumptions in Appendix A, where we also present a more general version of the model.

The equilibrium of the model can be characterized by a system of first-order conditions and market-clearing conditions. In the interior solution, the marginal benefit of increasing the treatment variable equals the marginal cost, yielding an implicit equation for the optimal level of the treatment variable as a function of exogenous parameters and state variables. We log-linearize this equation around the steady state to obtain the estimating equation that motivates our empirical specification.

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The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the

identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

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The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

Before proceeding, we note several contributions relative to the closest existing work. While prior studies have documented a correlation between the treatment variable and the outcome variable in various contexts, our paper is among the first to provide credibly causal estimates using an instrumental variables strategy. Moreover, our analysis of mechanisms goes beyond what has been possible in earlier studies, shedding light on the channels through which the treatment variable affects the outcome variable.

3..2 Stability and Strategy-Proofness

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on market design by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in market design is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and

falsification exercises.

$$(i, s) \text{ has justified envy if } s \succ_i \mu(i) \text{ and priority}_{is} > \text{priority}_{js} \text{ for some } j \in \mu(s) \tag{4}$$

$$exists succ_i' neq succ_i : mu_{Boston}(succ_i', succ_{-i}) succ_i mu_{Boston}(succ_i, succ_{-i}) \tag{5}$$

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3..3 Welfare Criteria and Mechanism Comparison

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$$W_U(\mu) = \sum_{i=1}^{n} u_i(\mu(i)), \quad u_i(s) = \beta_1 q_s + \beta_2 d_{is} + \varepsilon_{is}, \quad \#\{\text{matchings}\} = \binom{n+K-1}{K} \cdot \prod_{s=1}^{K} \lfloor q_s \rfloor! \tag{6}$$

$$W_R(\mu) = \min_{i=1,\ldots,n} u_i(\mu(i)) \tag{7}$$

$$P_{is} = \Pr(\mu(i) = s) = \Pr\Big(\text{priority}_{is} \geq \text{priority}_{\lceil q_s \rceil\text{-th}}\Big), \quad \begin{bmatrix} P_{i1} \\ P_{i2} \\ \vdots \\ P_{iK} \end{bmatrix} \in [0,1]^K, \ \sum_{s=1}^{K} P_{is} \leq 1 \tag{8}$$

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide

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Table 3: Welfare Comparison of School Assignment Mechanisms

	Stable?	Strategy-proof?	Avg Welfare	Worst-case Welfare
Student-proposing DA	Yes	Yes	0.742	0.484
School-proposing DA	Yes	No	0.731	0.412
Boston mechanism	No	No	0.718	0.314
Top Trading Cycles (TTC)	No	Yes	0.761	0.498
Random serial dictatorship	No	Yes	0.724	0.441
Immediate acceptance (IA)	No	No	0.706	0.287

The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4

percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

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4. Data

Our primary data source is administrative records, which covers 50,000 individuals over the period 1990–2010. This dataset provides detailed information on the outcome variable, the treatment variable, and a rich set of control variables that we use to address potential confounders. The data are available at the individual level and include both repeated cross-sections and a longitudinal component that allows us to track individuals over time. We discuss the advantages and limitations of this dataset relative to alternatives in the literature.

Table **??** presents summary statistics for the key variables used in our analysis. The mean of the outcome variable is consistent with values reported in prior studies, providing confidence that our sample is representative of the population of interest. The standard deviation suggests substantial variation across individuals, which is essential for our identification strategy. We also report the 10th and 90th percentiles to characterize the full distribution and to assess the extent to which our results might be driven by outliers in the tails.

We construct our main sample by merging several administrative and survey datasets. After

applying standard sample restrictions and dropping observations with missing values on key variables, our final estimation sample contains 50,000 observations covering 1990–2010. The merge rate across datasets exceeds 95 percent, alleviating concerns about selective attrition at the matching stage. We verify that the merged sample is representative by comparing moments of the key variables to those in the underlying source datasets.

The key independent variable, the treatment variable, exhibits substantial cross-sectional variation. The interquartile range spans a factor of three, providing sufficient variation to identify the effect of interest. We verify that this variation is not driven by outliers or measurement error by conducting a series of diagnostic tests. In particular, we show that the results are robust to winsorizing the treatment variable at the 1st and 99th percentiles, to using alternative measures from different data sources, and to correcting for measurement error using an errors-in-variables estimator.

We supplement the main dataset with auxiliary sources to construct additional control variables and instruments. Geographic controls come from standard sources in the literature, including distance to the coast, latitude, rainfall, temperature, and terrain ruggedness. Demographic controls are measured at baseline to avoid bad control problems. Institutional controls are drawn from widely used cross-country datasets. We lag all time-varying controls by one period to mitigate simultaneity concerns.

The measurement of the outcome variable deserves discussion. Following the standard approach in the literature, we use the logarithm of the outcome variable as our primary outcome variable. This transformation has three advantages: it reduces the influence of outliers, it allows us to interpret coefficients as approximate percentage changes, and it is consistent with the multiplicative structure of the theoretical model. We verify that our results are qualitatively robust to using the outcome variable in levels or in ranks.

Panel A of Table **??** reports summary statistics for the full sample, while Panels B and C restrict attention to subsamples that will be used in the heterogeneity analysis. The key variables are well-balanced across subsamples, with the exception of the treatment variable, which exhibits the expected variation driven by the quasi-experimental design. We formally test for balance across subsamples using a multivariate F-test and fail to reject the null of joint equality at conventional significance levels.

Data quality is a potential concern in studies of this type. We address this concern in several ways. First, we cross-validate our primary measure of the outcome variable against alternative sources and find high concordance rates. Second, we examine the time series of key variables for breaks or anomalies that might indicate measurement problems. Third, we re-estimate our main specifications on a subsample that excludes observations flagged as potentially problematic by the data providers. All results are robust to these checks.

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Table 4: Distribution of Student Preferences

	1 school	2 schools	3-5 schools	6-10 schools	11+ schools
Fraction of applicants (%)	12.4	18.7	31.4	28.8	8.7
Mean test score of 1st choice (std. dev.)	0.12	0.18	0.24	0.31	0.28
Fraction assigned to 1st choice (%)	84.1	72.4	61.8	54.2	49.8
Fraction assigned to any listed school (%)	84.1	88.3	91.4	93.8	95.2
Distance from 1st choice school (km)	1.84	2.14	2.74	3.12	3.84

Notes: Students who list more schools are more likely to be assigned to any listed school. Maximum list length is 12 schools. About 8.7% of applicants list 11+ schools and are nearly always assigned to a listed school.

Table 5: DA Algorithm Assignment Outcomes

	2005-2008	2009-2012	2013-2016	2017-2018
Assigned to 1st choice (%)	61.4	63.8	66.2	68.4
Assigned to 2nd choice (%)	14.2	13.8	12.4	11.8
Assigned to 3rd+ choice (%)	11.8	10.4	9.8	9.1
Unassigned (%)	12.6	12.0	11.6	10.7
Justified envy rate (%)	0.0	0.0	0.0	0.0
Avg. rank of assigned school	1.74	1.71	1.68	1.63

Notes: Cohorts split into 4-year periods. Justified envy rate = fraction of student-school pairs where student i prefers school s and has higher priority than an assigned student. DA produces stable matchings so this is exactly zero by construction.

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4..1 Student and School Characteristics

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on market design by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from

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The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

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$$Y_i = \alpha + \tau D_i + f(R_i - c) + \varepsilon_i, \quad D_i = \mathbf{1}[R_i \geq c] \tag{9}$$

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A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

Before proceeding, we note several contributions relative to the closest existing work. While prior studies have documented a correlation between the treatment variable and the outcome variable in various contexts, our paper is among the first to provide credibly causal estimates using an instrumental variables strategy. Moreover, our analysis of mechanisms goes beyond what has been possible in earlier studies, shedding light on the channels through which the treatment variable affects the outcome variable.

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variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on market design by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

5..1 Regression Discontinuity at Admission Cutoffs

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The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in market design is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

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Table 6: RDD First Stage: Probability of Attending School at Admission Cutoff

	(1)	(2)	(3)	(4)
Dep. var.: Attending school s in grade 6				
Above cutoff (sharp RD)	0.61***	0.61***	0.62***	0.62***
	((0.02))	((0.02))	((0.02))	((0.02))
Running variable (score - cutoff)	0.04***	0.04***	0.03***	0.03***
	((0.01))	((0.01))	((0.01))	((0.01))
Above x Running	-0.02*	-0.02*	-0.01	-0.01
	((0.01))	((0.01))	((0.01))	((0.01))
Baseline covariates	No	Yes	Yes	Yes
School-year FE	No	No	Yes	Yes
Bandwidth (points)	10	10	10	8
Observations	24,841	24,841	24,841	21,382
Mean dep. var. (below cutoff)	0.12	0.12	0.12	0.12

Notes: *** p<0.01, ** p<0.05, * p<0.1. Running variable is the normalized priority score minus cutoff. Sharp RD: above cutoff indicator. Bandwidth selected by Calonico-Cattaneo-Titiunik (2014) MSE-optimal procedure. Standard errors clustered by school-cohort.

The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

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6. Results

Table **??** presents our main estimates. Column (1) reports the baseline specification without controls. Columns (2) through (4) progressively add demographic controls, geographic controls, and fixed effects. The point estimate remains remarkably stable across specifications, suggesting that selection on unobservables is unlikely to drive our results. Following the bounding approach of Oster (2019), we show that the degree of selection on unobservables required to explain away our estimate would need to be substantially larger than the degree of selection on observables.

The estimated coefficient on the treatment variable is statistically significant at the one percent level in all specifications. The magnitude implies that a one standard deviation increase in the

treatment variable is associated with a positive and significant of approximately 0.3 standard deviations in the outcome variable. To put this in perspective, this effect size is comparable to the impact of an additional year of schooling on earnings, one of the most well-established causal relationships in economics. The 95 percent confidence interval is sufficiently narrow to rule out effects smaller than one-tenth of a standard deviation.

We find no evidence that our results are driven by pre-existing trends or differential selection. The event study coefficients are small and statistically insignificant in all pre-treatment periods, lending support to our parallel trends assumption. Moreover, the treatment effect emerges sharply at the time of the intervention rather than gradually, which is consistent with a causal interpretation and inconsistent with pre-existing trends that happen to accelerate at the time of treatment.

The first-stage relationship between the proposed instrument and the treatment variable is strong, with an F-statistic well above conventional thresholds for weak instruments. The first-stage coefficient is precisely estimated and has the expected sign. Following the recommendations of Andrews, Stock, and Sun (2019), we also report the effective F-statistic and the tF critical values. Under all weak-instrument diagnostics considered, we can reject the null of a weak first stage at the 5 percent level.

Heterogeneity analysis reveals that the treatment effect is concentrated among individuals with below-median baseline levels of the outcome variable. This pattern is consistent with the theoretical prediction that the marginal return to the treatment variable is decreasing. We also find suggestive evidence of heterogeneity along demographic dimensions, although the subsample estimates are less precisely estimated due to smaller sample sizes. The difference in treatment effects across subgroups is statistically significant at the 10 percent level.

We complement the regression analysis with non-parametric evidence on the treatment effect. Kernel density estimates of the distribution of the outcome variable show a clear rightward shift for treated individuals relative to untreated individuals. Quantile treatment effects, reported in the appendix, are positive and statistically significant throughout the distribution, with somewhat larger effects in the lower tail. This pattern suggests that the treatment reduces inequality in the outcome variable by disproportionately benefiting those at the bottom of the distribution.

The magnitude of our estimates is informative about the relative importance of the treatment variable as a determinant of the outcome variable. A variance decomposition exercise reveals

that variation in the treatment variable can account for approximately 15 to 25 percent of the cross-sectional variance in the outcome variable, depending on the specification. While this leaves substantial residual variation unexplained, it establishes the treatment variable as one of the most quantitatively important determinants identified in the literature.

We also examine the dynamics of the treatment effect by interacting the treatment variable with indicators for different time horizons. The effect is small and statistically insignificant in the first year after treatment, grows steadily over the subsequent three years, and stabilizes at a new long-run level by year four. This gradual emergence of the treatment effect is consistent with the theoretical model, which predicts that the adjustment to the new equilibrium occurs over multiple periods as agents update their behavior and as general equilibrium effects propagate through the economy.

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The first-stage relationship between the proposed instrument and the treatment variable is

Table 7: RDD Estimates: Effect of Attending Oversubscribed School on 8th Grade Test Scores

	(1)	(2)	(3)	(4)
Panel A: Math scores (standard deviations)				
ITT (above cutoff)	0.18***	0.19***	0.19***	0.18***
	((0.05))	((0.05))	((0.05))	((0.06))
IV (2SLS: attend = 1)	0.30***	0.31***	0.31***	0.29***
	((0.09))	((0.09))	((0.08))	((0.10))
Panel B: Reading scores (standard deviations)				
ITT (above cutoff)	0.12**	0.13**	0.13**	0.12**
	((0.05))	((0.05))	((0.05))	((0.06))
IV (2SLS: attend = 1)	0.20**	0.21**	0.21**	0.19**
	((0.09))	((0.09))	((0.08))	((0.10))
Baseline covariates	No	Yes	Yes	Yes
School-year FE	No	No	Yes	Yes
Bandwidth (points)	10	10	10	8
Observations	22,184	22,184	22,184	19,013
First-stage F	841	841	847	798

Notes: *** p<0.01, ** p<0.05, * p<0.1. ITT = intent-to-treat. IV uses above-cutoff indicator as instrument for school attendance. Standard errors clustered by school-cohort. Scores normalized to mean 0, std. dev. 1 within grade-year.

Table 8: RDD Estimates: Effect of Attending Oversubscribed School on High School Outcomes

	(1)	(2)	(3)	(4)
Panel A: High school graduation (pp)				
ITT (above cutoff)	4.2***	4.4***	4.3***	4.1***
	((1.4))	((1.4))	((1.4))	((1.5))
IV (2SLS)	6.9***	7.2***	6.9***	6.6***
	((2.3))	((2.3))	((2.3))	((2.5))
Panel B: 4-year college enrollment (pp)				
ITT (above cutoff)	3.1**	3.4**	3.3**	3.0**
	((1.4))	((1.4))	((1.4))	((1.5))
IV (2SLS)	5.1**	5.6**	5.3**	4.8**
	((2.3))	((2.3))	((2.3))	((2.5))
Baseline covariates	No	Yes	Yes	Yes
School-year FE	No	No	Yes	Yes
Bandwidth (points)	10	10	10	8
Observations	18,421	18,421	18,421	15,847
Mean dep. var. (below cutoff, %)	74.2	74.2	74.2	74.2

Notes: *** p<0.01, ** p<0.05, * p<0.1. Outcome is an indicator for the event. IV uses above-cutoff indicator as instrument.

Table 9: Heterogeneous Effects of Attending Oversubscribed School by Student Subgroup

	Math (sd)	Reading (sd)	Graduation (pp)	College (pp)
All students (IV baseline)	0.30***	0.20**	6.9***	5.1**
	((0.09))	((0.09))	((2.3))	((2.3))
Free/reduced lunch	0.34***	0.23**	8.1***	6.2**
	((0.11))	((0.11))	((2.8))	((2.8))
Not free/reduced lunch	0.24**	0.14	5.2*	3.4
	((0.12))	((0.13))	((3.0))	((3.1))
English language learner	0.41***	0.28**	9.4***	7.1**
	((0.14))	((0.14))	((3.4))	((3.4))
Special education	0.22	0.18	4.1	2.8
	((0.18))	((0.18))	((4.2))	((4.3))
Female	0.28***	0.22**	6.4**	6.2**
	((0.10))	((0.10))	((2.6))	((2.6))
Male	0.32***	0.18*	7.4***	4.1*
	((0.10))	((0.10))	((2.7))	((2.5))

Notes: *** p<0.01, ** p<0.05, * p<0.1. IV estimates using RDD at admission cutoffs. Each row estimated separately with baseline covariates and school-year FE. Standard errors clustered by school-cohort.

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6..1 Test Score Outcomes

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The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

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6..2 High School Graduation and College Enrollment

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The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

7. Welfare Analysis

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on market design by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional

variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in market design is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

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A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

Table 10: Boston Mechanism vs. Deferred Acceptance: Outcomes Under Misreported Preferences

	Truthful	Strategize top choice	Strategize full list	Nash equilibrium
Panel A: Boston mechanism				
Assigned to 1st true preference (%)	71.4	68.4	61.8	58.4
Avg welfare (0-1 scale)	0.718	0.694	0.672	0.651
Welfare loss from manipulation (pp)	0.0	2.4	4.6	6.7
Panel B: Student-proposing DA				
Assigned to 1st true preference (%)	66.2	66.2	66.2	66.2
Avg welfare (0-1 scale)	0.742	0.742	0.742	0.742
Welfare loss from manipulation (pp)	0.0	0.0	0.0	0.0

Notes: Welfare measured as fraction of first-choice school expected value captured. Boston mechanism exhibits welfare reducing manipulation in equilibrium (6.7 pp loss). DA is strategy-proof so manipulation has no effect.

Before proceeding, we note several contributions relative to the closest existing work. While prior studies have documented a correlation between the treatment variable and the outcome variable in various contexts, our paper is among the first to provide credibly causal estimates using an instrumental variables strategy. Moreover, our analysis of mechanisms goes beyond what has been possible in earlier studies, shedding light on the channels through which the treatment variable affects the outcome variable.

Table 11: TTC vs. DA: Assignment Differences and Welfare

	TTC	DA	TTC-DA	p-value
Assigned to 1st choice (%)	68.4	66.2	2.2	0.01
Assigned to any listed school (%)	94.1	91.8	2.3	0.00
Average rank of assigned school	1.61	1.68	-0.07	0.00
Utilitarian welfare	0.761	0.742	0.019	0.00
Rawlsian welfare (min utility)	0.498	0.484	0.014	0.04
Stable matchings achieved (%)	N/A	100	–	–
Justified envy (% of pairs)	1.84	0.00	1.84	0.00

Notes: TTC uses current assignment as endowment. DA = student-proposing DA. Welfare calculated from estimated preference parameters. Justified envy under TTC: 1.84% of student-school pairs where a student prefers a school over their TTC assignment and has higher priority than an assigned student.

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on market design by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

7..1 Counterfactual Mechanism Comparison

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A central question in market design is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

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7..2 Distributional Welfare Effects

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

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8. Conclusion

This paper provides new evidence on the causal effect of the treatment variable on the outcome variable. Using an instrumental variables strategy applied to administrative records, we find a positive and significant and statistically significant effect that is robust across a wide range of specifications. The magnitude of the effect is economically meaningful and has clear implications for both theory and policy.

Our findings have several implications for economic theory. First, the results support models in which the treatment variable plays a central role in determining the outcome variable, as opposed to alternative theories that emphasize other factors. Second, the pattern of heterogeneity in treatment effects is consistent with the predictions of the theoretical framework, providing a rare example of a model that is both internally consistent and empirically validated. Third, the dynamics of the treatment effect shed light on the speed of adjustment to economic shocks, a parameter that is notoriously difficult to estimate.

From a policy perspective, the estimates suggest that interventions targeting the treatment variable could have meaningful effects on the outcome variable. However, several caveats are in order. First, our estimates are local to the population and context studied, and external validity to other settings should not be assumed without additional evidence. Second, our identification strategy relies on assumptions that, while supported by the evidence, cannot be directly tested. Third, the general equilibrium effects of large-scale policy interventions may differ from the partial equilibrium effects captured by our estimates.

Future work should investigate the mechanisms through which the treatment variable affects the outcome variable in greater detail. Our results identify a reduced-form relationship but do not fully pin down the structural channels at work. Complementary evidence from randomized controlled trials, structural estimation, or natural experiments in other contexts would strengthen

the conclusions drawn here. Additionally, understanding the long-run persistence of the effects documented in this paper is an important question for future research.

In conclusion, this paper demonstrates that the treatment variable is a quantitatively important determinant of the outcome variable. The combination of a compelling identification strategy, rich data, and robust results across specifications provides a strong foundation for the causal claims made here. We hope that these findings will stimulate further research on the economic consequences of variation in the treatment variable and will inform the design of policies aimed at improving the outcome variable.

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9. Appendix A: Matching Theory Proofs

Consider an economy populated by a continuum of agents indexed by $i \in [0, 1]$. Each agent chooses the outcome variable to maximize lifetime utility subject to a budget constraint. The key source of heterogeneity across agents is their exposure to the treatment variable. We solve for the equilibrium of this economy and derive comparative statics with respect to the key parameters. The model generates a clear testable prediction: an exogenous increase in the treatment variable should lead to a positive and significant change in the outcome variable, with the magnitude depending on structural parameters that we estimate in the empirical analysis.

Let Y_i denote the outcome of interest for unit i. We model Y_i as a function of treatment status D_i, observed covariates X_i, and unobserved heterogeneity ε_i. Under standard regularity conditions, the conditional expectation function takes the form presented in Equation **??**. The parameter of interest is the coefficient on treatment status, which captures the causal effect of the treatment variable on the outcome variable holding constant observed and unobserved confounders. Identification of this parameter requires an exclusion restriction, which we discuss in detail below.

The theoretical framework builds on the canonical model of field. Agents face a discrete choice between J alternatives, each yielding utility $U_{ij} = V_{ij} + \epsilon_{ij}$, where V_{ij} is the deterministic component and ϵ_{ij} captures idiosyncratic preferences. Under standard distributional assumptions on the error term, the choice probabilities take a tractable closed form. The model can be estimated using maximum likelihood or method of moments, and the resulting structural parameters can be

used to simulate counterfactual policy scenarios.

We derive the key prediction of the model through a simple comparative statics exercise. An exogenous increase in the treatment variable shifts the equilibrium allocation of the outcome variable through two channels: a direct effect operating through prices, and an indirect effect operating through the reallocation of factors across sectors. The total effect is ambiguous in general, but under the empirically relevant parameter values, we show that the direct effect dominates. Proposition 1 in Appendix A formalizes this result and establishes conditions under which the sign of the total effect is unambiguous.

Under the assumption that the proposed instrument satisfies the exclusion restriction, the model implies that the reduced-form relationship between the proposed instrument and the outcome variable is proportional to the structural parameter of interest. This motivates our instrumental variables strategy described in Section **??**. A key advantage of the IV approach is that it does not require functional form assumptions on the relationship between the treatment variable and the unobserved confounders. The cost is that we identify a local average treatment effect rather than the population average treatment effect.

The model delivers additional testable implications beyond the main prediction. First, the treatment effect should be larger for individuals with lower baseline levels of the outcome variable, reflecting diminishing marginal returns. Second, the effect should be persistent over time rather than transitory, because the mechanism operates through permanent changes in the level of the treatment variable. Third, the effect should be concentrated in sectors with greater exposure to the underlying channel. We test each of these auxiliary predictions in Section 6.

A.1 Proof that DA Produces a Stable Matching

Proposition 1 (Stability of DA)**.** *The student-proposing Deferred Acceptance algorithm produces a stable matching.*

Proof. Suppose for contradiction that (i, s) is a blocking pair in the DA outcome μ_D. Then $s \succ_i \mu_D(i)$, so student i proposed to school s at some step t before proposing to $\mu_D(i)$. Since $i \notin \mu_D(s)$, school s must have rejected i at some step $t' \geq t$ in favor of students with higher priority. Let j be a student in $\mu_D(s)$; then $\text{priority}_{js} > \text{priority}_{is}$ (since s rejected i for j). This contradicts the assumption that

$i \succ_s j$, so no blocking pair exists and μ_D is stable. □

A.2 Proof of Strategy-Proofness of Student-Proposing DA

Proposition 2 (Strategy-Proofness)**.** *Under student-proposing DA, truthful preference reporting is a dominant strategy for every student.*

Proof. Fix the reports of all other students $\succ_{-i}$. We show that for any misreport $\succ'_i$, $\mu_D(\succ_i, \succ_{-i}) \succsim_i \mu_D(\succ'_i, \succ_{-i})$.

Case 1: $\mu_D(\succ_i, \succ_{-i}) = \mu_D(\succ'_i, \succ_{-i})$. Then the student is indifferent.

Case 2: $\mu_D(\succ'_i, \succ_{-i}) \succ_i \mu_D(\succ_i, \succ_{-i})$. Let $s^* = \mu_D(\succ'_i, \succ_{-i})$ and $s = \mu_D(\succ_i, \succ_{-i})$. Under truthful play, i proposed to s^* before s. School s^* rejected i. Under the misreport, i is matched to s^*; but this requires s^* to hold i throughout. Since the set of other students' proposals is the same in both runs (they report $\succ_{-i}$ in both), the students competing for s^* are identical. Therefore s^* holds i under the misreport iff it holds i under truthful play – a contradiction. So Case 2 is impossible. □

A.3 Proof that Boston Mechanism is Manipulable

Proposition 3 (Boston Manipulability)**.** *The Boston mechanism is not strategy-proof: there exist preference profiles $(\succ_i, \succ_{-i})$ and misreports $\succ'_i$ such that $\mu_B(\succ'_i, \succ_{-i}) \succ_i \mu_B(\succ_i, \succ_{-i})$.*

Proof by example. Consider two students $\{i, j\}$ and two schools $\{s, t\}$ each with capacity 1. True preferences: $i : s \succ t$, $j : s \succ t$. Priorities: s ranks j above i. Under truthful play, both apply to s first; s accepts j and rejects i; i applies to t and is accepted. Under misreport $\succ'_i$: $t \succ s$, student i applies to t first (no competition), gets t; student j gets s. The outcomes are identical. Now consider: true preferences $i : s \succ t$, $j : t \succ s$. Under truthful play: i gets s (no competition), j gets t. Under misreport $i' : t \succ s$: both i and j apply to t; t accepts j (higher priority); i applies to s, gets s. Again the same.

For a case where manipulation strictly helps: let true preferences be $i : s \succ t$, $j : s \succ t$, priority of s: $j > i$. Under truthful play i ends up at t. Now if i knows j will apply to s first-round, i can safely report $\succ'_i$: $s \succ t$ truncated to only s – but this is the same as truthful play. Alternatively, if there are three schools and i can misreport to avoid competing at s, obtain their second choice at a school with no competition first-round, and j does not fill it. This straightforward construction

establishes that reporting preferences strategically can yield a weakly better assignment, with strict improvement possible. □

A.4 Rural Hospitals Theorem

[Rural Hospitals Theorem] In any market with two-sided preferences, the set of agents matched and the number of positions filled at each hospital (school) are the same across all stable matchings.

Proof. Let μ and μ' be two stable matchings. For any school s, suppose $|\mu(s)| > |\mu'(s)|$. Then there exists a student $i \in \mu(s) \setminus \mu'(s)$. Under μ', i is either unmatched or matched to $\mu'(i) \prec_i s$. In either case, (i, s) is a blocking pair for μ' (since s prefers i to some student in $\mu'(s)$ by priority consistency), contradicting the stability of μ'. Hence $|\mu(s)| = |\mu'(s)|$ for all s. □

A.5 Combinatorial Bounds on the Number of Stable Matchings

Lemma 1 (Counting Stable Matchings)**.** *The number of distinct stable matchings satisfies:*

$$|\mathcal{M}_{stable}| \leq \binom{n+K-1}{K} \cdot \prod_{s=1}^{K} \binom{n_s}{\lfloor q_s \rfloor}, \tag{10}$$

where n_s is the number of students who rank school s and $q_s = \lceil c_s \rceil$ is the school capacity. The matching probability matrix is:

$$\mathbf{P} = \begin{bmatrix} P_{11} & P_{12} & \cdots & P_{1K} \\ P_{21} & P_{22} & \cdots & P_{2K} \\ \vdots & \vdots & \ddots & \vdots \\ P_{n1} & P_{n2} & \cdots & P_{nK} \end{bmatrix}, \quad \sum_{s=1}^{K} P_{is} \leq 1, \ \sum_{i=1}^{n} P_{is} \leq \lfloor q_s \rfloor. \tag{11}$$

We make several simplifying assumptions to maintain tractability. First, we assume that agents have rational expectations and perfect information about the distribution of shocks. Second, we abstract from general equilibrium effects by treating prices as exogenous to individual agents. Third, we assume that the production technology exhibits constant returns to scale. We discuss the robustness of our theoretical predictions to relaxing each of these assumptions in Appendix A, where we also present a more general version of the model.

The equilibrium of the model can be characterized by a system of first-order conditions and market-clearing conditions. In the interior solution, the marginal benefit of increasing the treatment variable equals the marginal cost, yielding an implicit equation for the optimal level of the treatment variable as a function of exogenous parameters and state variables. We log-linearize this equation around the steady state to obtain the estimating equation that motivates our empirical specification.

10. Appendix B: DA Algorithm Details

This appendix provides additional results referenced in the main text. We organize the supplementary material into three parts. Appendix A presents proofs of the theoretical propositions stated in Section **??**. Appendix B reports additional empirical results, including alternative specifications and sample definitions. Appendix C describes the construction of key variables in detail.

We present detailed derivations of the key theoretical results stated in Section **??**. The proofs proceed under standard regularity conditions that are verified in the data. We state these conditions formally as Assumptions A1 through A4 below, and provide discussion of their economic content and empirical plausibility. Throughout, we maintain the notation introduced in the main text.

This section reports results from additional robustness exercises not included in the main text due to space constraints. All estimates are qualitatively similar to the baseline results reported in Table **??**. We organize these results by the type of robustness check: alternative control variables, alternative sample restrictions, alternative functional forms, alternative estimation methods, and alternative definitions of key variables.

We provide here a detailed description of the data sources and variable construction procedures. For each variable, we report the original source, the years available, the unit of observation, and any transformations applied. We also report summary statistics for the full set of variables used in the analysis, including those that appear only in robustness checks or auxiliary regressions.

The following tables report the complete set of coefficient estimates for all control variables included in our main specifications. In the interest of space, the main text reports only the coefficients on the key independent variables. The full results presented here confirm that the control variables enter with expected signs and magnitudes, and that their inclusion does not materially affect the coefficients of interest.

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11. Appendix C: Additional Results

Table **??** presents our main estimates. Column (1) reports the baseline specification without controls. Columns (2) through (4) progressively add demographic controls, geographic controls, and fixed effects. The point estimate remains remarkably stable across specifications, suggesting that selection on unobservables is unlikely to drive our results. Following the bounding approach of Oster (2019), we show that the degree of selection on unobservables required to explain away our estimate would need to be substantially larger than the degree of selection on observables.

The estimated coefficient on the treatment variable is statistically significant at the one percent level in all specifications. The magnitude implies that a one standard deviation increase in the treatment variable is associated with a positive and significant of approximately 0.3 standard deviations in the outcome variable. To put this in perspective, this effect size is comparable to the impact of an additional year of schooling on earnings, one of the most well-established causal relationships in economics. The 95 percent confidence interval is sufficiently narrow to rule out effects smaller than one-tenth of a standard deviation.

We find no evidence that our results are driven by pre-existing trends or differential selection. The event study coefficients are small and statistically insignificant in all pre-treatment periods, lending support to our parallel trends assumption. Moreover, the treatment effect emerges sharply at the time of the intervention rather than gradually, which is consistent with a causal interpretation and inconsistent with pre-existing trends that happen to accelerate at the time of treatment.

The first-stage relationship between the proposed instrument and the treatment variable is strong, with an F-statistic well above conventional thresholds for weak instruments. The first-stage coefficient is precisely estimated and has the expected sign. Following the recommendations of Andrews, Stock, and Sun (2019), we also report the effective F-statistic and the tF critical values. Under all weak-instrument diagnostics considered, we can reject the null of a weak first stage at

the 5 percent level.

Heterogeneity analysis reveals that the treatment effect is concentrated among individuals with below-median baseline levels of the outcome variable. This pattern is consistent with the theoretical prediction that the marginal return to the treatment variable is decreasing. We also find suggestive evidence of heterogeneity along demographic dimensions, although the subsample estimates are less precisely estimated due to smaller sample sizes. The difference in treatment effects across subgroups is statistically significant at the 10 percent level.

Table 12: Appendix: Estimated Preference Parameters from Structural Demand Model

	Coeff.	Std. Error	95% CI Low	95% CI High
School test score (1 sd increase)	1.84***	0.21	1.43	2.25
Distance (1 km increase)	-0.42***	0.08	-0.58	-0.26
Peer free lunch share (10 pp increase)	-0.18**	0.09	-0.36	-0.00
School size (100 students)	0.08*	0.04	-0.00	0.16
Teacher experience share (10 pp)	0.24***	0.07	0.10	0.38
Fraction same race as student	0.31***	0.08	0.15	0.47

Notes: *** p<0.01, ** p<0.05, * p<0.1. Preference parameters estimated by BLP (1995) method using variation in school availability as instruments. Standard errors from 200 bootstrap replications.

We complement the regression analysis with non-parametric evidence on the treatment effect. Kernel density estimates of the distribution of the outcome variable show a clear rightward shift for treated individuals relative to untreated individuals. Quantile treatment effects, reported in the appendix, are positive and statistically significant throughout the distribution, with somewhat larger effects in the lower tail. This pattern suggests that the treatment reduces inequality in the outcome variable by disproportionately benefiting those at the bottom of the distribution.

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